Electric Potential and Electric Potential Energy Due to Point Charges

7.5

In the first section of this chapter, you saw how a Van de Graaff generator in a science museum causes the hair of anyone in contact with the device to stand on end. At that point, the discussion dealt simply with the properties of electric charge, and how the like charges (electrons) on individual hairs caused the hairs to repel each other and spread out in all directions.

We can look at this phenomenon in another way: the charge on the Van de Graaff generator creates a high electric potential on the conducting sphere of the generator. This potential provides an electric field that exerts a force on charged particles near the generator. This force, in turn, accelerates these particles. In fact, when the Van de Graaff generator was invented, it was an early type of particle accelerator. As a particle accelerator, however, the Van de Graaff generator has limitations. For one thing, it can only accelerate charges like those on the sphere. Also, if the potential on the sphere becomes too high with respect to the generator's surroundings, the air around the generator will ionize. This causes large sparks to jump between the sphere and objects at lower electric potentials, resulting in artificial lightning (**Figure 1**).

At a distance, the conducting sphere of a Van de Graaff generator resembles a point charge. In this section, you will learn about electric potentials, electric potential energies, and electric potential differences for point charges and their surroundings.

Figure 1 The charge on the Van de Graaff generator has a large electric potential difference with its surroundings. This can cause large sparks to appear between the generator and its surroundings.

Electric Potential Due to a Point Charge

Earlier you learned about the electric properties of a point charge. This simple example is useful in many situations. In Section 7.4, we defined electric potential V in terms of a test charge q and the electric potential energy E_E. The electric potential energy, in turn, depends on the position d of the test charge in an electric field E. If the charge producing the electric field has the same sign as the test charge q, the work done by the electric field on q is negative. This means that $\Delta E_E > 0$ for $\Delta d > 0$, and the sign of potential energy (and thus potential) is reversed from that of the uniform electric field.

For a point charge q, the magnitude of the electric field is

$$E = \frac{kq}{r^2}$$

Setting the location in the field equal to the radius of the field ($\Delta d = r$), the **electric potential due to a point charge** is inversely proportional to the distance from the charge and directly proportional to the amount of charge. So, for a point charge q producing an electric field, at a distance r from the charge, the electric potential is

$$V = \frac{kq\Delta d}{r^2}$$

$$V = \frac{kq}{r}$$

electric potential due to a point charge the electric potential is inversely proportional to the distance from the charge and proportional to the amount of charge producing the field

These results reinforce what you learned about potential energy and the behaviour of like and unlike charges.

If the charge producing $\vec{E}$ has the same sign as the test charge, the work done on q by the electric field is negative.

↳ $\Delta E_E > 0$, $\Delta d > 0$, sign of potential and potential energy is reversed from that of $\vec{E}$.

$$V = \frac{kq\Delta d}{r^2}$$

$$V = \frac{kq}{r}$$

E = High potential energy → low poten^l energy

the source of electrical energy, a positive test charge will naturally move from the Electric Potential Energy Due to Point Charges 355

positive terminal where a high potential exists to the negative terminal where a low potential exists. Since the electric field points from positive to negative, the positive test charge will also move in the same direction as the field. Conversely, electrons will naturally travel from a region of low potential to a region of high potential, in a direction opposite to the direction of the electric field.

Suppose you place a source charge ($q > 0$) at the origin of the coordinate system in **Figure 2**. The charge produces a potential given by $\frac{kq}{r}$. Then you place a positive test charge nearby. Both the source and the test charge are positive, so they repel one another, and the test charge experiences a force that carries it "downhill" and away from the origin along the blue potential curve in Figure 2. When the source charge is negative, the potential is described by the lower, red curve in Figure 2. The source charge is now negative, so the attractive force carries a positive test charge along the red potential curve toward the origin.

Figure 2 Electric potential near a point charge. The blue curve is for a positive point charge, and the red curve is for a negative point charge.

$V = q > 0$

q (+)

$V = q < 0$

Only changes in potential energy are important, so you must always be clear about choosing the reference point for potential energy. Electric potential is proportional to electric potential energy, so you must also pay attention to the reference point for electric potential. For example, when dealing with a point charge, follow the standard convention by choosing $V = 0$ at an infinite distance from the source charge. Infinite distance represents a distance so far away that any potential produced by a charge is negligible and for all intents and purposes equal to zero. In many other problems, choose Earth as $V = 0$ because Earth conducts charge well. Put another way, the electric ground is the point where $V = 0$.

In finite distance: $V = 0$ at an infinite distance from a source charge.

For two or more charges, use the superposition principle. Remember that electric potentials of negative charges are negative, and electric potentials of positive charges are positive. However, because electric potential is not a vector, the total electric potential of several point charges equals the algebraic sum of the electric potentials resulting from each of the individual charges and their specific distances from the chosen location.

Electric potentials of negative charges are negative and vice-versa. But when finding total electric charge, use the algebraical sum, (ignore direction)

Electric Potential Energy of Two Point Charges

Now consider the electric potential energy of a pair of charges. The electric potential energy of a pair of charges is the potential energy possessed by each charge in the pair because of its position. For a pair of charges, this potential energy is inversely proportional to the distance between the charges and directly proportional to the product of the two charges. For a pair of charges q_1 and q_2 separated by a distance r, use the equation for the electric field for the point charge q_1 and the definition of electric potential energy for the point charge q_2 in terms of V to determine the potential energy for the two charges:

$$E_E = q_2 V$$

$$= \frac{kq_2}{r} q_1$$

$$E_E = \frac{kq_1q_2}{r}$$

The electric potential energy of a pair of charges is then given by:

$$E_E = \frac{kq_1q_2}{r}$$

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$$E_E = \frac{kq_1q_2}{r}$$

$$\Delta E_E = E_{Ef} - E_{Ei}$$

$$\Delta E_E = \frac{kq_1q_2}{r_f} - \frac{kq_1q_2}{r_i}$$

$$= kq_1q_2\left(\frac{1}{r_f} - \frac{1}{r_i}\right)$$

For 2 like charges, ΔE_E is +ve and increases as they are brought together.
For 2 unlike charges, ΔE_E is (−ve) and decreases as it is an attractive force

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This is qualitatively similar to the equation for Coulomb's law, except that E_E varies as $\frac{1}{r}$, not $\frac{1}{r^2}$. If the charges are initially separated by a distance r_i and then brought together to a final separation r_f, the change in the potential energy is

$$\Delta E_E = E_{Ef} - E_{Ei}$$

$$\Delta E_E = \frac{kq_1q_2}{r_f} - \frac{kq_1q_2}{r_i}$$

Note that only changes in potential energy matter, and that E_E approaches zero when the two charges are very far apart. The blue curve in **Figure 3** shows the general behaviour of the electric potential energy for two like charges. In this case, E_E is positive and increases as the charges are brought together. If the charges have opposite signs, the electric force is attractive (negative), and the potential energy is then also negative, as shown by the red curve in Figure 3. The following Tutorial will demonstrate how to solve problems involving electric potential and electric potential energy.

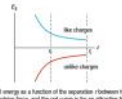

Figure 3 Electric potential energy as a function of the separation r between two charges q_1 and q_2. The blue curve is for a repulsive force, and the red curve is for an attractive force.

$E_{Ef} > E_{Ei}$

$\Delta E_E = E_{Ef} - E_{Ei}$

$\Delta E_E = \frac{kq_1q_2}{r_f} - \frac{kq_1q_2}{r_i}$

Both like charges are brought together (+ve) $\Delta E_E > 0$ gain PE

Unlike charges Attract $\Delta E_E < 0$ loss PE

$-4 - -2 = (-2)$

$E_E = 0$

$E_E = 0$

Tutorial 1 Electric Potential and Electric Potential Energy

Sample Problem 1 models how to calculate the total electric potential of two point charges and the work required to move a third charge into this potential. In Sample Problem 2, we will use the electric potential energy and conservation of energy to calculate the speed of a charge at a given location. Finally, in Sample Problem 3, we show how to calculate the minimum particle separation by using electric potential energy, conservation of energy, and conservation of momentum.

Sample Problem 1: Calculating the Electric Potential

A point charge with a charge of 4.00×10^{-8} C is 4.00 m due west from a second point charge with a charge of -1.00×10^{-7} C.

(a) Calculate the total electric potential due to these charges at a point P, 4.00 m due north of the first charge.

(b) Calculate the work required to bring a third point charge with a charge of 2.0×10^{-9} C from infinity to point P.

Solution

(a) **Given:** $q_1 = 4.00 \times 10^{-8}$ C; $q_2 = -1.00 \times 10^{-7}$ C; $r_{1P} = 4.00$ m; $r_{2P} = 4.00$ m

Required: V_T

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A point charge with a charge of 4.00×10^{-8} C is 4.00 m due west from a second point charge with a charge of -1.00×10^{-7} C.

(a) Calculate the total electric potential due to these charges at a point P, 4.00 m due north of the first charge.

(b) Calculate the work required to bring a third point charge with a charge of 2.0×10^{-9} C from infinity to point P.

1a)

$V_T = ?$

$V = \frac{kq}{r}$

$V_1 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{4}$

$V_1 = 89.9$ J/C

$V_2 = ?$

$r_{2P} = \sqrt{(r_{1P})^2 + (r_{12})^2}$

$= \sqrt{(4)^2 + (4)^2}$

$= \sqrt{32}$

$r_{2P} = 5.6569$ m

$V_2 = \frac{(8.99 \times 10^9)(-1 \times 10^{-7})}{5.6569}$

$V_2 = -158.92$

Analysis: The total potential at P is the sum of the potentials due to the two individual charges q_1 and q_2. We can use the equation $V = \frac{kq}{r}$ to calculate the potential due to each charge and the Pythagorean theorem to calculate the distance of point P from q_2: $r_{2P} = \sqrt{r_{1P}^2 + r_{12}^2}$.

Solution: V_T is the sum of the potentials due to the individual charges q_1 and q_2. Calculate the potential V_1 at P due to q_1:

$$V_1 = \frac{kq_1}{r_{1P}}$$

$$= \frac{\left(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)(+4.00 \times 10^{-8}\text{ C})}{4.00\text{ m}}$$

$$= 8.99 \times 10^2\text{ N}\cdot\text{m/C}$$

$$V_1 = 8.99 \times 10^2\text{ J/C}$$

To calculate the potential V_2 at P due to q_2, first calculate the distance r_{2P} between P and q_2:

$$r_{2P} = \sqrt{r_{1P}^2 + r_{12}^2}$$

$$r_{2P} = \sqrt{(4.00\text{ m})^2 + (4.00\text{ m})^2}$$

$$r_{2P} = 5.6569\text{ m (two extra digits carried)}$$

Then calculate V_2:

$$V_2 = \frac{kq_2}{r_{2P}}$$

$$= \frac{\left(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)(-1.00 \times 10^{-7}\text{ C})}{5.6569\text{ m}}$$

$$= -1.5892 \times 10^2\text{ N}\cdot\text{m/C}$$

$$V_2 = -1.5892 \times 10^2\text{ J/C (two extra digits carried)}$$

Now calculate the total potential V_T:

$$V_T = V_1 + V_2$$

$$= 8.99 \times 10^2\text{ J/C} + (-1.5892 \times 10^2\text{ J/C})$$

$$V_T = -68.92\text{ J/C (one extra digit carried)}$$

Statement: The total electric potential at point P is −68.9 J/C.

$$V_2 = \frac{(8.99\times10^9)(-1\times10^{-7})}{5.6569}$$

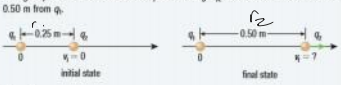
$V_2 = -158.92$

$V_T = 89.9 + (-158.92) = \boxed{-69.02\ J/C}$

b)

Sample Problem 2: Electric Potential Energy and Dynamics

A point charge q_1 with charge 2.0×10^{-6} C is initially at rest at a distance of 0.25 m from a second charge q_2 with charge 8.0×10^{-6} C and mass 4.0×10^{-9} kg (**Figure 4**). Both charges are positive. Charge q_1 remains fixed at the origin, whereas q_2 travels to the right upon release. Determine the speed of charge q_2 when it reaches a distance of 0.50 m from q_1.



$E_{E1} = \frac{kq_1q_2}{r_1}$ $\quad E_{K1} = 0$

$\Sigma E_1 = E_{E1} + 0$

$\Sigma E_2 = E_{E1}$

$E_{E1} = E_{E2} + E_{K2}$

$\frac{kq_1q_2}{r_1} = \frac{kq_1q_2}{r_2} + \frac{mv^2}{2}$

$\frac{kq_1q_2}{r_1} = \frac{kq_1q_2}{2r_1} + \frac{mv^2}{2}$

$\frac{kq_1q_2}{r_1} - \frac{kq_1q_2}{2r_1} = \frac{mv^2}{2}$

$\frac{k_1q_1q_2}{r_1}(1-\frac{1}{2}) = \frac{mv^2}{2}$

$\frac{k_1q_1q_2}{r_1}(\frac{1}{2})$

$\frac{k_1q_1q_2}{2r_1} = \frac{mv^2}{2}$

$v^2 = \frac{2k_1q_1q_2}{2r_1m}$

$v = \sqrt{\frac{k_1q_1q_2}{r_1m}}$

$v = \sqrt{\frac{(8.99\times10^9)(2\times10^{-6})(8\times10^{-6})}{(0.25)(4\times10^{-9})}}$

$v = 1.2\times10^4$

1).

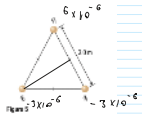


Figure 5

q_1 1.5 1.5 q_3

6×10^{-6} $\quad -3\times10^{-6}$

$V_{q_1} = \frac{kq_1}{r} = \frac{8.99\times10^9(6\times10^{-6})}{1.5} = 33\ 960$

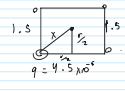
$V_{q_3} = \frac{kq_3}{r} = \frac{8.99\times10^9(-3\times10^{-6})}{1.5} = -17980$

V_{q_2}

$x = \sqrt{3^2 - 1.5^2}$

$x = 2.598$

2)



$x = \sqrt{(\frac{1.5}{2})^2 + (\frac{1.5}{2})^2}$

$x = 1.0606$

$V = \frac{kq}{r}$

$V = \frac{(8.99\times10^9)(4.5\times10^{-8})}{1.0606}$

$V = 38141.339$

$U = 38141\times4 = 152565.359 = 1.5\times10^5$ J/C

3. Two electrons start at rest with a separation of 5.0×10^{-12} m. Once released, the electrons accelerate away from each other. Calculate the speed of each electron when they are a very large distance apart. (ans: 7.1×10^6 m/s)

$r = \infty = \frac{1}{\infty}$

5.0×10^{-12} m

$E_E = \frac{kq_1q_2}{r} = \frac{(8.99\times10^9)(-1.6\times10^{-19})^2}{(5.0\times10^{-12})}$

$= 4.6029\times10^{-17}$

$4.60\times10^{-17} = \frac{mv^2}{2}$

$4.6\times10 = 9.11\times10^{-31}\ v^2$

$V = 7.10\times10^6$

1 a) lower potential energy and lower electric potential

b) lower potential energy but higher electric potential.

2). Two particles with the same V dont necessarily have the same E_E. Exif the charge is different

3) If the E_E stays the same, no work is done

4) $E_E = \frac{Kq_1q_2}{r}$

$\frac{\sin 2x}{\ldots} = \tan^3 x$

2) Two particles with the same V don't necessarily have the same E_E. Even if the charge is different

3) If the E_E stays the same, no work is done.

4) $E_E = \frac{Kq_1q_2}{r}$

$r = \frac{Kq_1q_2}{E_E}$

$= \frac{8.99\times10^9 (9.5\times10^{-5})(8.5\times10^{-6})}{40}$

$= 0.85\,m$

5) $\Delta E_E = \frac{Kq_1q_2}{r_f} - \frac{Kq_1q_2}{r_i}$

$= Kq_1q_2\left(\frac{1}{r_f} - \frac{1}{r_i}\right)$

$\Delta E_E = (8.99\times10^9)(9.5\times10^{-5})(8.5\times10^{-6})\left(\frac{1}{1.5} - \frac{1}{0.3}\right)$

$\Delta E_E = 9.2\,J$

$$\frac{\sin 2x}{\cot^2 x(1+\cos 2x)} = \tan^3 x$$

$$\left[\begin{array}{l}\sin 2x = 2\sin x\cos x \\ \cos 2x = \cos^2 x - \sin^2 x\end{array}\right]$$

$$= \frac{2\sin x\cos x}{\cot^2 x(1+\cos^2 x - \sin^2 x)}$$

$$\left[1 = \sin^2 x + \cos^2 x\right]$$

$$= \frac{2\sin x\cos x}{\cot^2 x(\cancel{\sin^2 x} + \cos^2 x + \cos^2 x - \cancel{\sin^2 x})}$$

$$= \frac{2\sin x\cos x}{\cot^2 x(2\cos^2 x)}$$

$$\left[\cot^2 x = \frac{\cos^2 x}{\sin^2 x}\right]$$

$$= \frac{2\sin x\cos x}{\frac{\cos^2 x}{\sin^2 x}\cdot 2\cos^2 x}$$

$$= \frac{2\sin x\cos x}{\frac{2\cos^4 x}{\sin^2 x}}$$

$$= \frac{2\sin x\cos x}{1} \div \frac{2\cos}{\sin^2 x}$$

$$= \frac{2\sin x\cos x}{1} \times \frac{\sin^2 x}{2\cos^4 x}$$

$$= \frac{\cancel{2}\sin^3 x\cos x}{\cancel{2}\cos^4 x}$$

$$= \frac{\sin^3 x\cdot\cancel{\cos x}}{(\cos^3 x)\cancel{\cos x}}$$

$$= \frac{\sin^3 x}{\cos^3 x}$$

$$= \tan^3 x$$

$a\log_p w = 3$ $\qquad$ $\log_w p = 2a^2$

$a\times\frac{\log w}{\log p} = 3$ $\qquad$ $\frac{\log p}{\log w} = 2a^2$

$\frac{a\log w}{\log p} = 3$

$\frac{\log w}{\log p} = \frac{3}{a}$

This means that:

$\frac{\log p}{\log w} = \frac{a}{3}$ and $\frac{\log p}{\log w} = 2a^2$

$\frac{a}{3}\times3 = 2a^2\times3$

$\frac{a}{a} = \frac{6a^2}{a}$

$\frac{1}{6} = \frac{6a}{6}$

$\boxed{\frac{1}{6} = a}$

↳ ax^3+bx^2+cx+d form

↳ odd function

↳ 3rd difference is -30 (or some stupid shit like that)

↳ 1 x-intercept at $x=2$

↳ cubic function

In vertex form: only one bracket, since only one x-int.

$f(x) = a(\quad)$

- cubic, so has order 3

$f(x) = a(\quad)^3$

- x int at $x=2$

$f(x) = a(x-2)^3$

- 3rd diff @ -30

f.d = a n!

$-30 = a(3)!$

$-30 = a(1\times2\times3)$

$-30 = a(6)$

vertex form: $-5(x-2)^3$

now to get ax^3+bx^2+cx+d factor

$-5(x-2)(x-2)(x-2)$

$(-5x+10)(x-2)(x-2)$

$(-5x^2+10x+10x-20)(x-2)$

$(-5x^2+20x-20)(x-2)$

$(-5x^3+20x^2-20x+10x^2-40x+40)$

$(-5x^3+30x^2-60x+40)$

$\boxed{-5x^3+30x^2-60x+40}$

$4x$

$-30 = a(3)!$

$-30 = a(1 \times 2 \times 3)$

$-30 = a(6)$

$\frac{-30}{6} = a$

$a = -5$

$(-5x^3 + 30x^2 - 60x + 40)$

$\boxed{-5x^3 + 30x^2 - 60x + 40}$